TIDY01: Fundamentals of Tidyverse I

Presented by Gabriel Rodrigues Palma

Tidyverse for Ecologists (TIDY01)

stats

Day 2 (9:30 - 17:30)

Data Cleaning
(9:35 - 11:05)

Grammar of Graphics
(13:30 - 15:00)

Data Filtering and
Selecting (11:05 - 12:30)

Grammar of Graphics

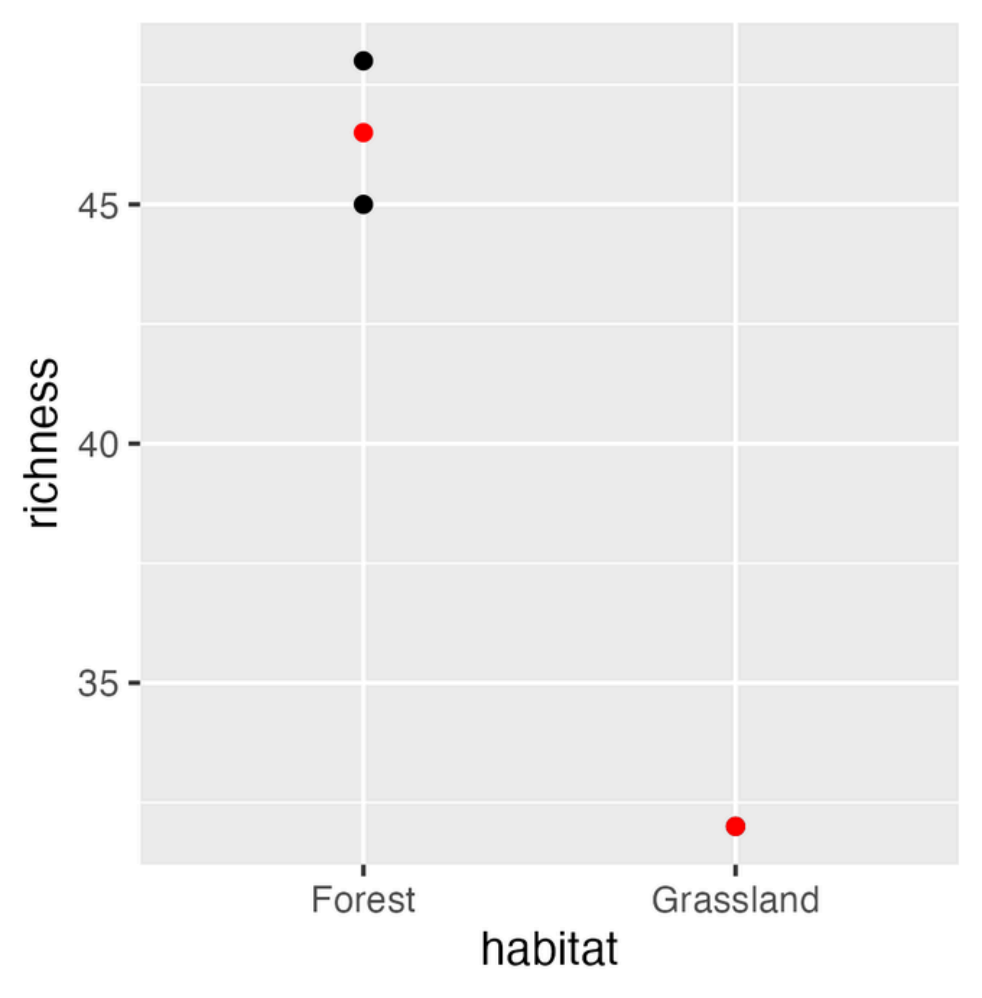
stats

Grammar of Graphics

R Input

```r
# Statistical summary example
biodiversity <- data.frame(
  habitat = c('Forest', 'Grassland', 'Forest'),
  richness = c(45, 32, 48)
)
ggplot(biodiversity, aes(x = habitat, y = richness)) +
  geom_point() + stat_summary(fun = mean, geom = 'point', color = 'red')
```

→

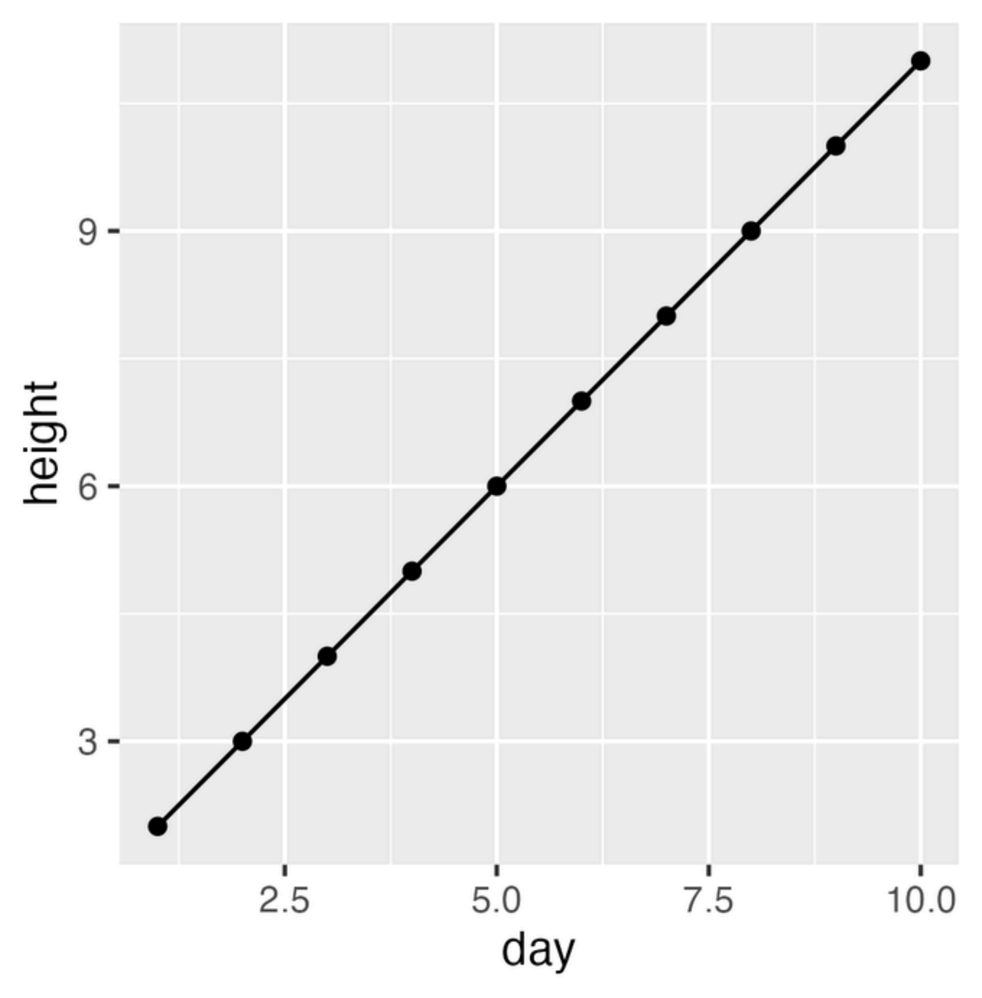


Grammar of Graphics

R Input

```r
# Multiple layers example
plant_growth <- data.frame(
  day = 1:10,
  height = c(2, 3, 4, 5, 6, 7, 8, 9, 10, 11)
)
ggplot(plant_growth, aes(x = day, y = height)) +
  geom_point() + geom_line()
```

→

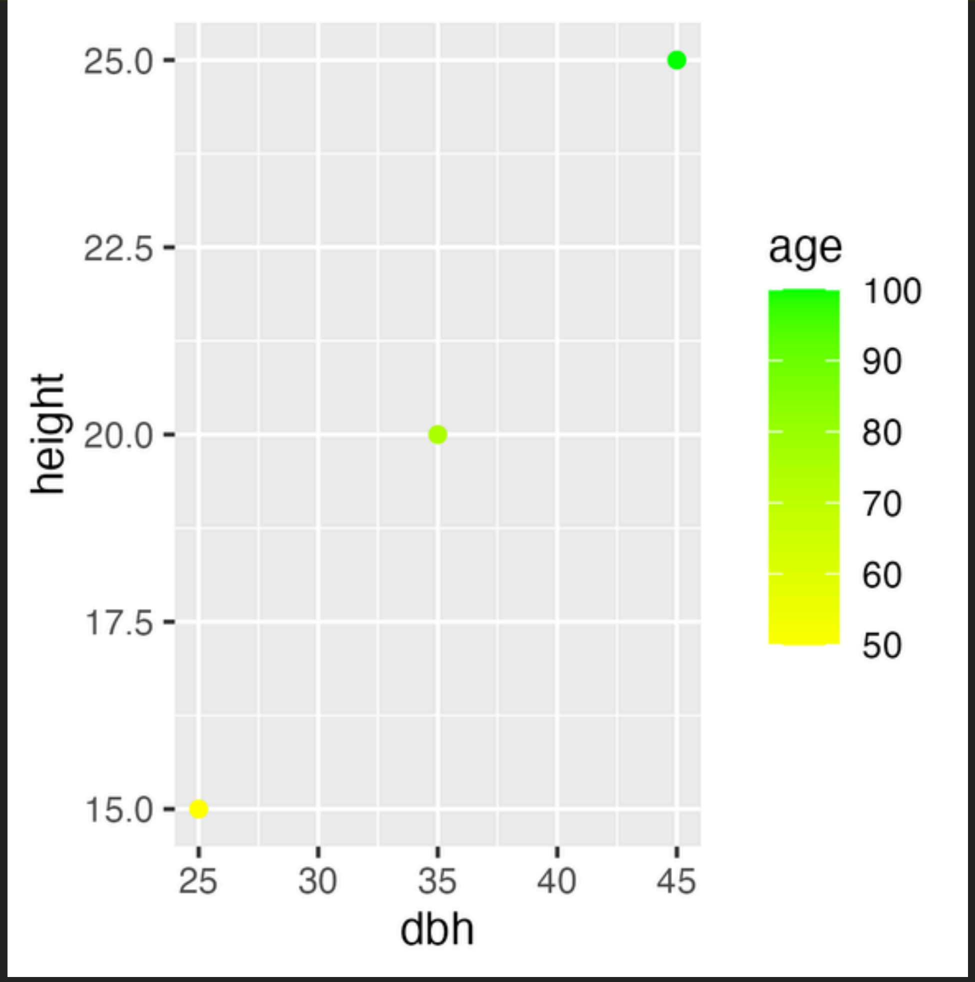


Grammar of Graphics

R Input

```r
# Custom scales example
tree_data <- data.frame(
  dbh = c(25, 35, 45),
  height = c(15, 20, 25),
  age = c(50, 75, 100)
)
ggplot(tree_data, aes(x = dbh, y = height, color = age)) +
  geom_point() + scale_color_gradient(low = 'yellow', high = 'green')
```

→

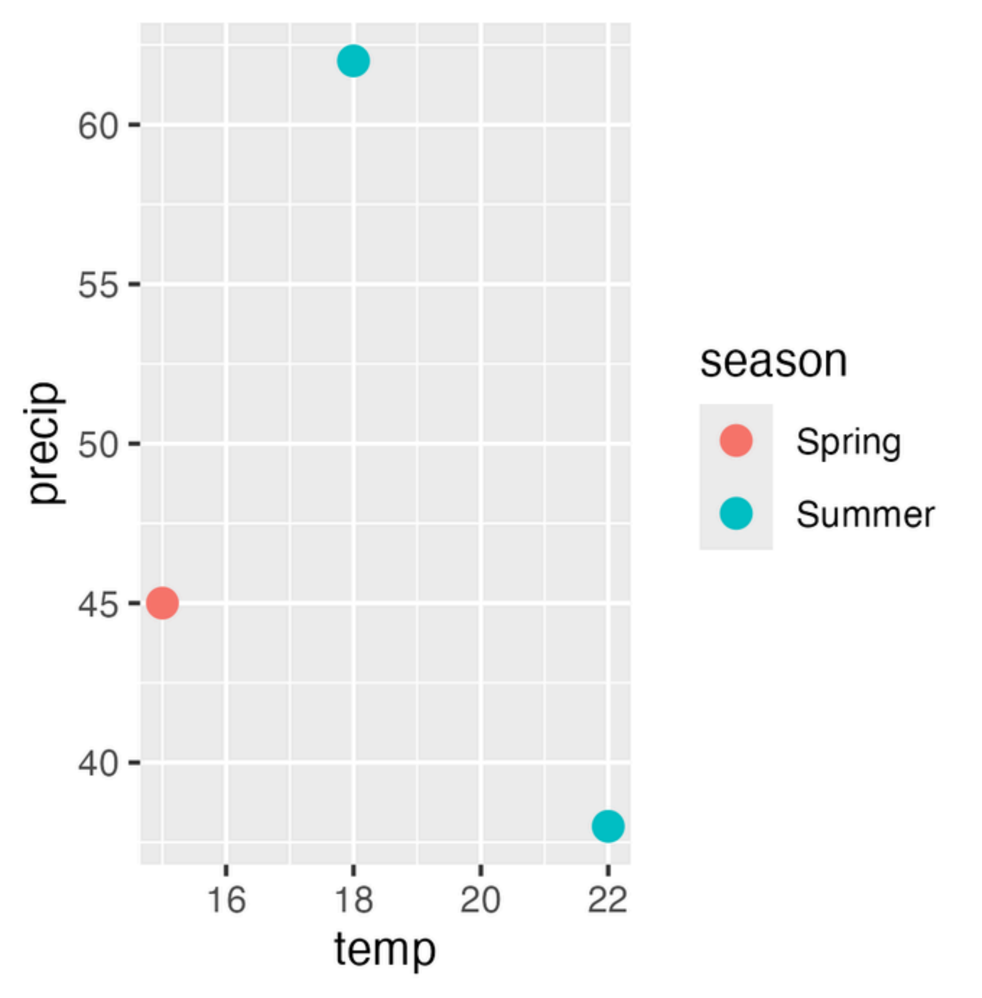


Grammar of Graphics

R Input

```r
# Adding color aesthetics
climate <- data.frame(
  temp = c(15, 18, 22),
  precip = c(45, 62, 38),
  season = c('Spring', 'Summer', 'Summer')
)
ggplot(climate, aes(x = temp, y = precip, color = season)) +
  geom_point(size = 3)
```

→

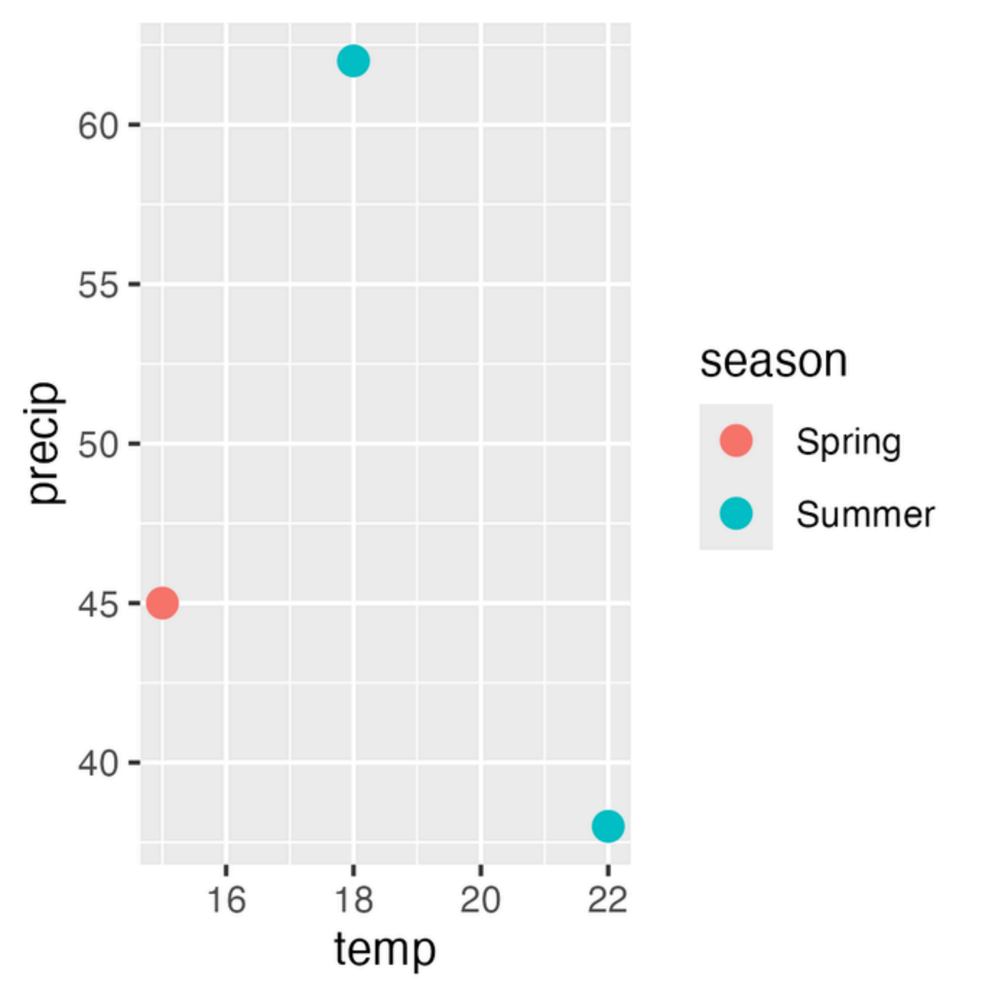


Grammar of Graphics

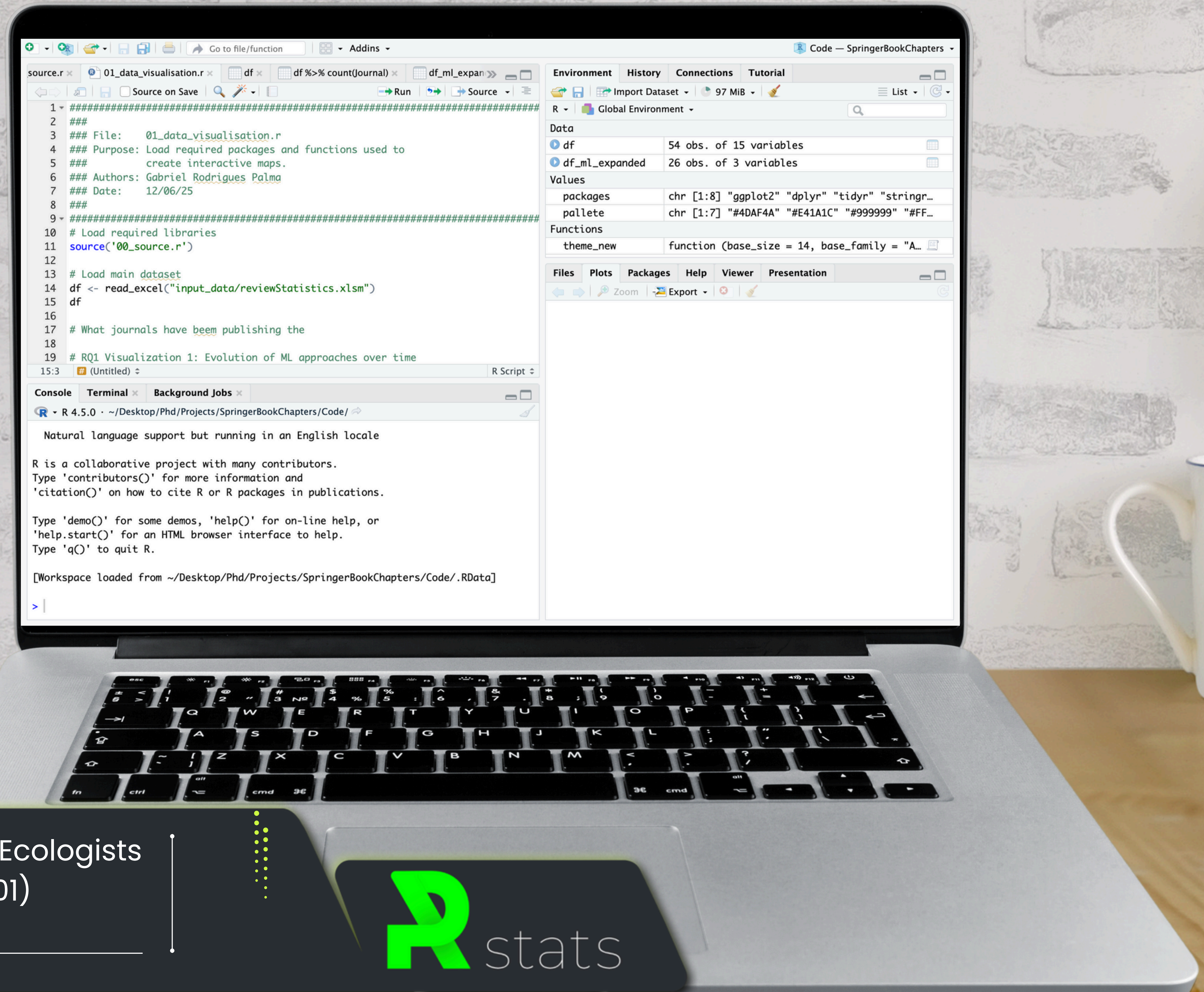
R Input

```r
# Basic ggplot structure
library(ggplot2)
species_data <- data.frame(
  species = c('Oak', 'Pine'),
  count = c(25, 18)
)
ggplot(species_data, aes(x = species, y = count)) +
  geom_point()
```

→



Grammar of Graphics



Tidyverse for Ecologists (TIDY01)

Rstats

Day 2 (9:30 - 17:30)

Data Cleaning
(9:35 - 11:05)

Data Filtering and Selecting (11:05 - 12:30)

Grammar of Graphics
(13:30 - 15:00)

Basic Plots
(15:00 - 16:30)

Basic Plots

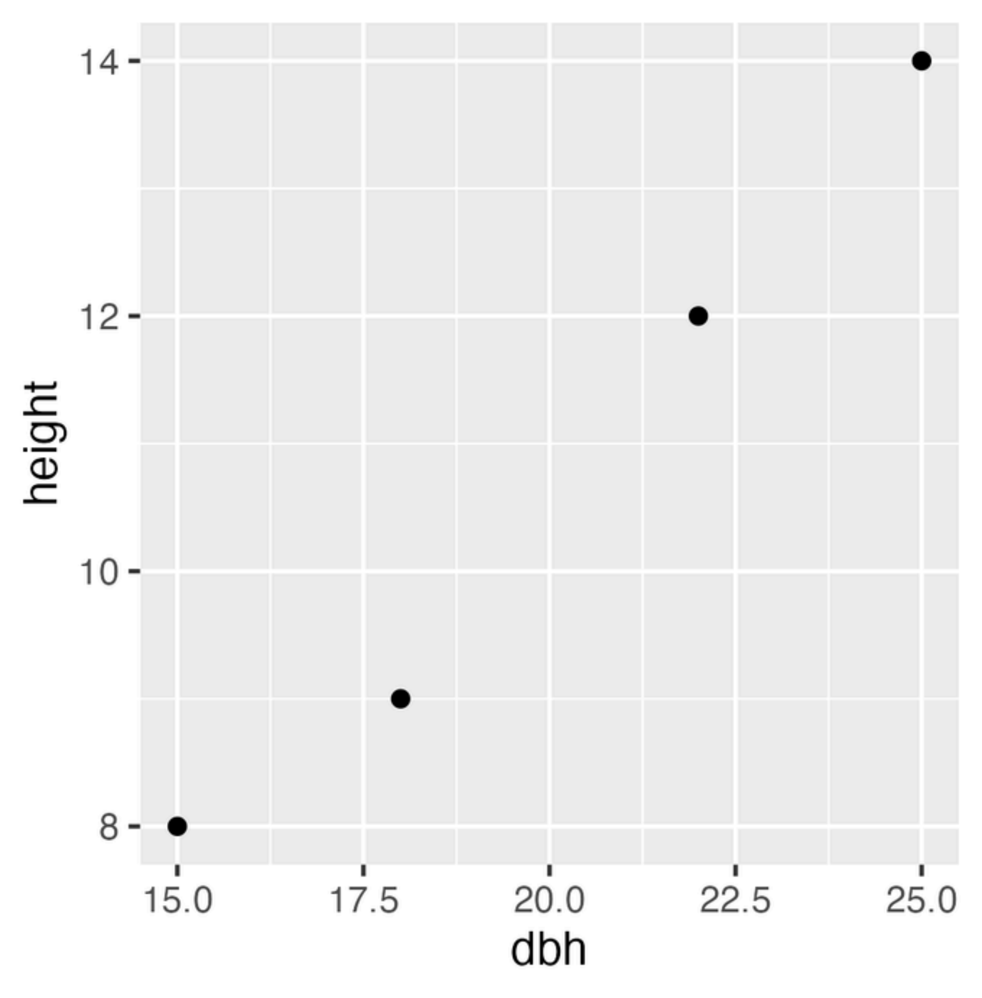

Total Confirmed
566,269
Confirmed Cases by
Country/Region/Sovereignty
92,932 US
China
NORTH
AMERICA

Basic Plots

R Input

```r
# Basic scatterplot
tree_data <- data.frame(
  dbh = c(15, 22, 18, 25),
  height = c(8, 12, 9, 14)
)
ggplot(tree_data, aes(x = dbh, y = height)) +
  geom_point()
```

→

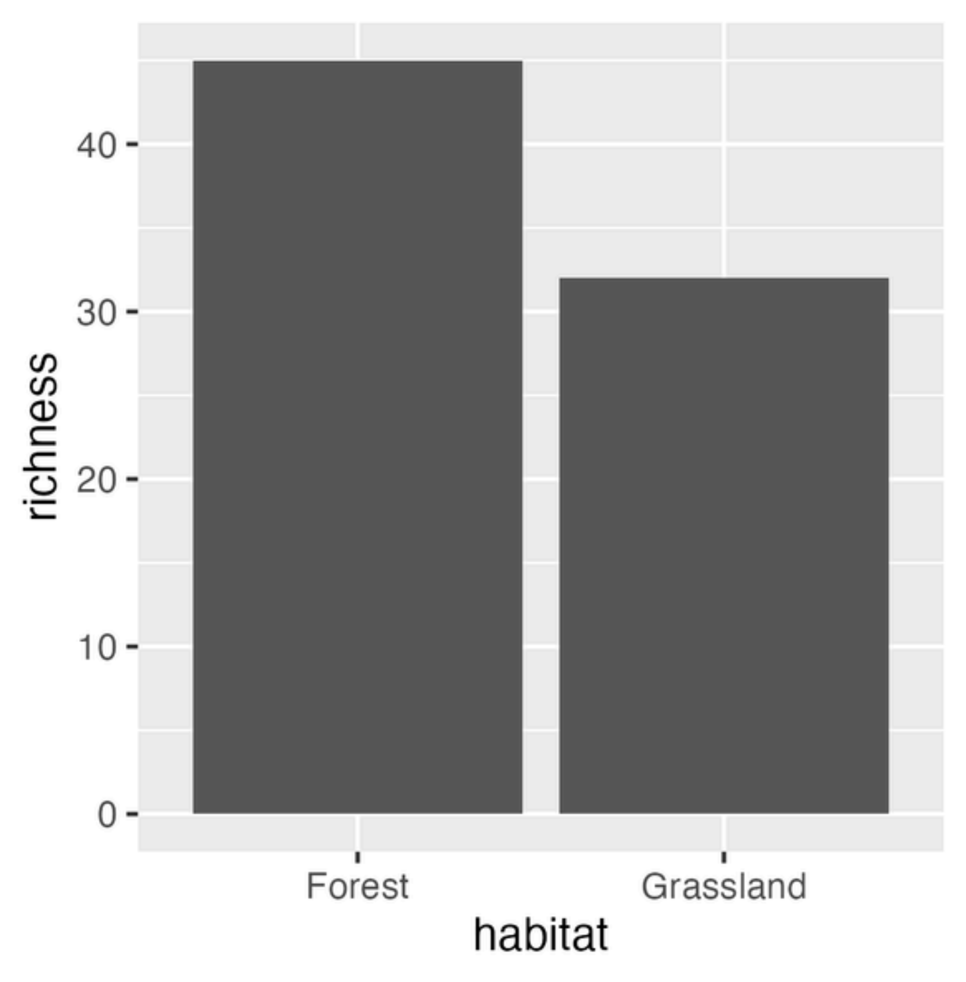


Basic Plots

R Input

```
# Bar chart with species counts
species_counts <- data.frame(
  habitat = c('Forest', 'Grassland'),
  richness = c(45, 32)
)
ggplot(species_counts, aes(x = habitat, y = richness)) +
  geom_col()
```

→

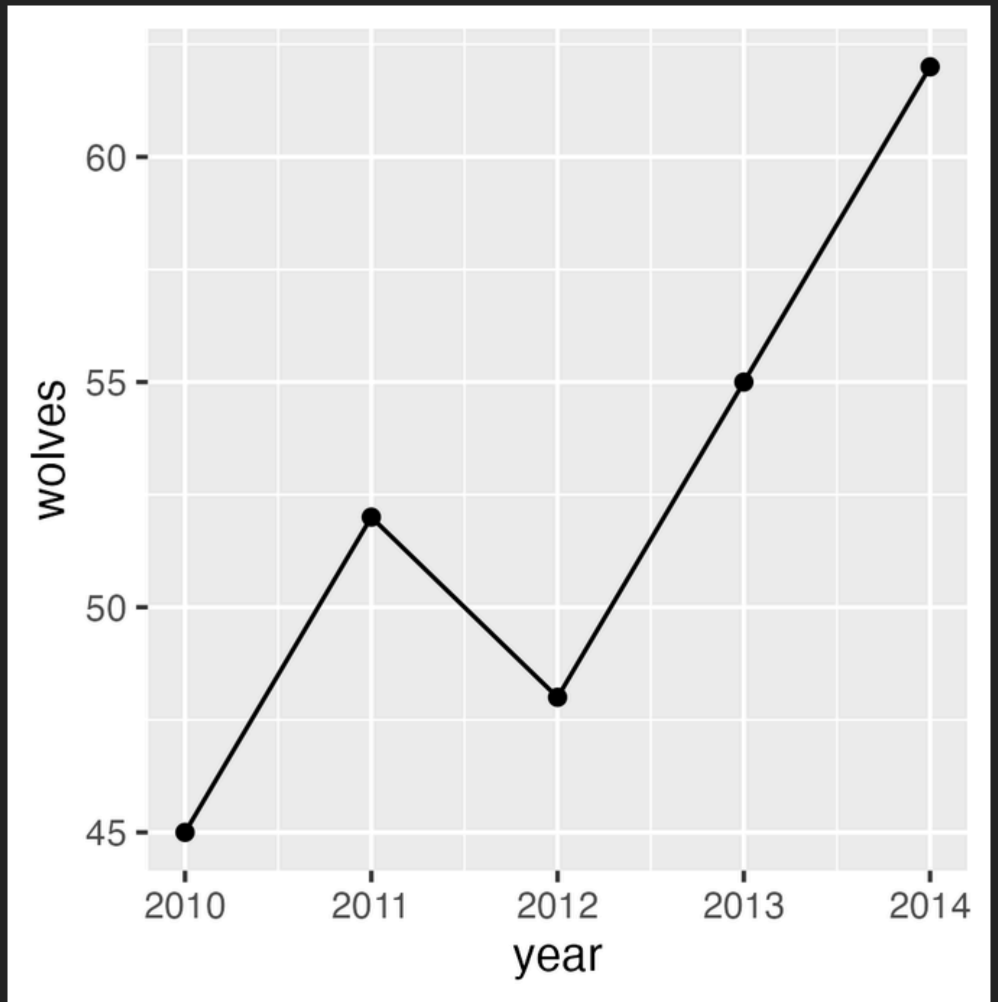


Basic Plots

R Input

```r
# Line graph for population trends
population <- data.frame(
  year = 2010:2014,
  wolves = c(45, 52, 48, 55, 62)
)
ggplot(population, aes(x = year, y = wolves)) +
  geom_line() + geom_point()
```

→

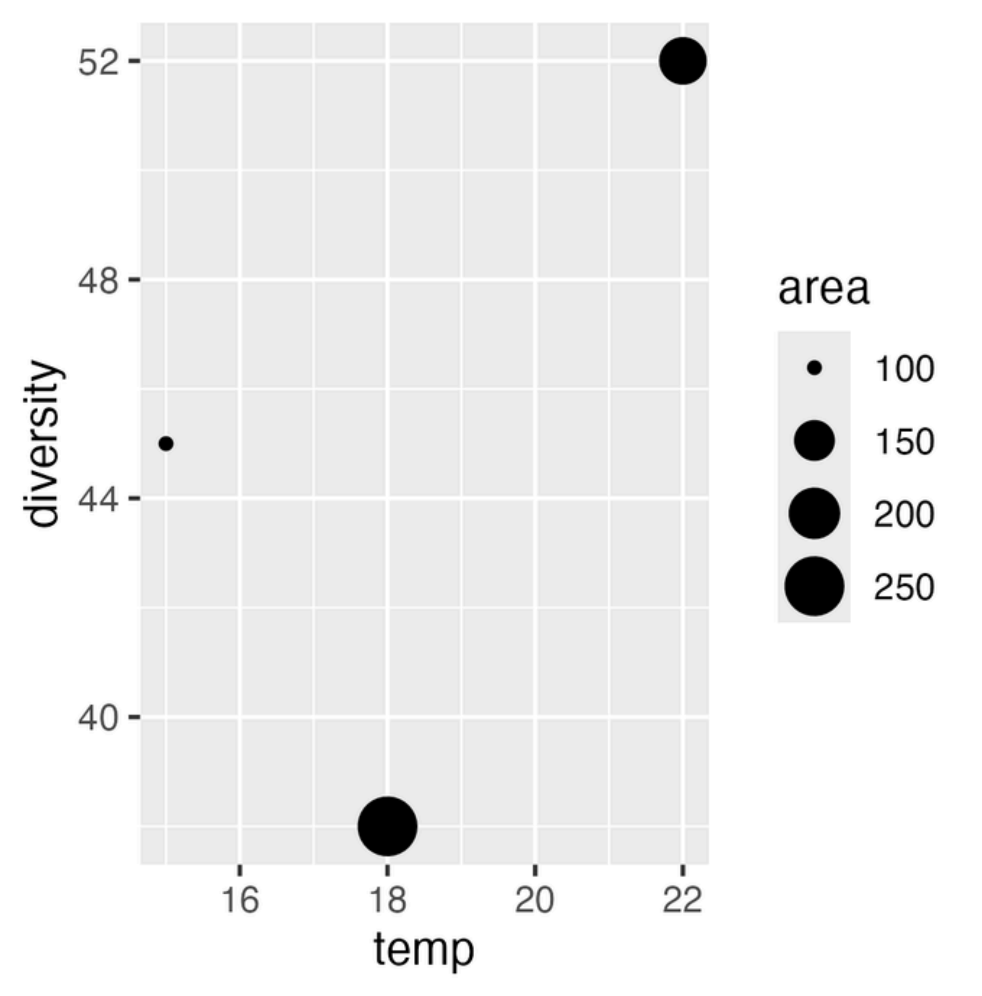


Basic Plots

R Input

```r
# Bubble plot with multiple variables
ecosystem <- data.frame(
  temp = c(15, 18, 22),
  diversity = c(45, 38, 52),
  area = c(100, 250, 180)
)
ggplot(ecosystem, aes(x = temp, y = diversity, size = area)) +
  geom_point()
```

→

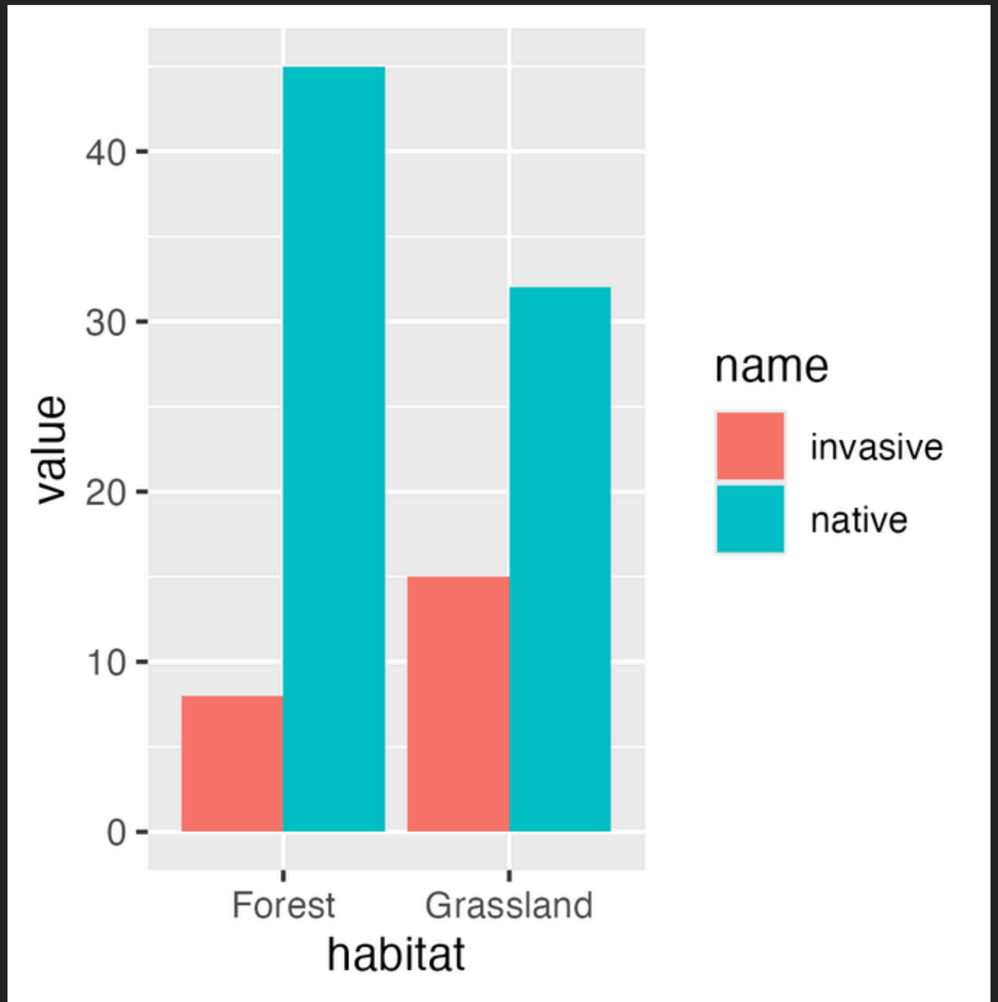


Basic Plots

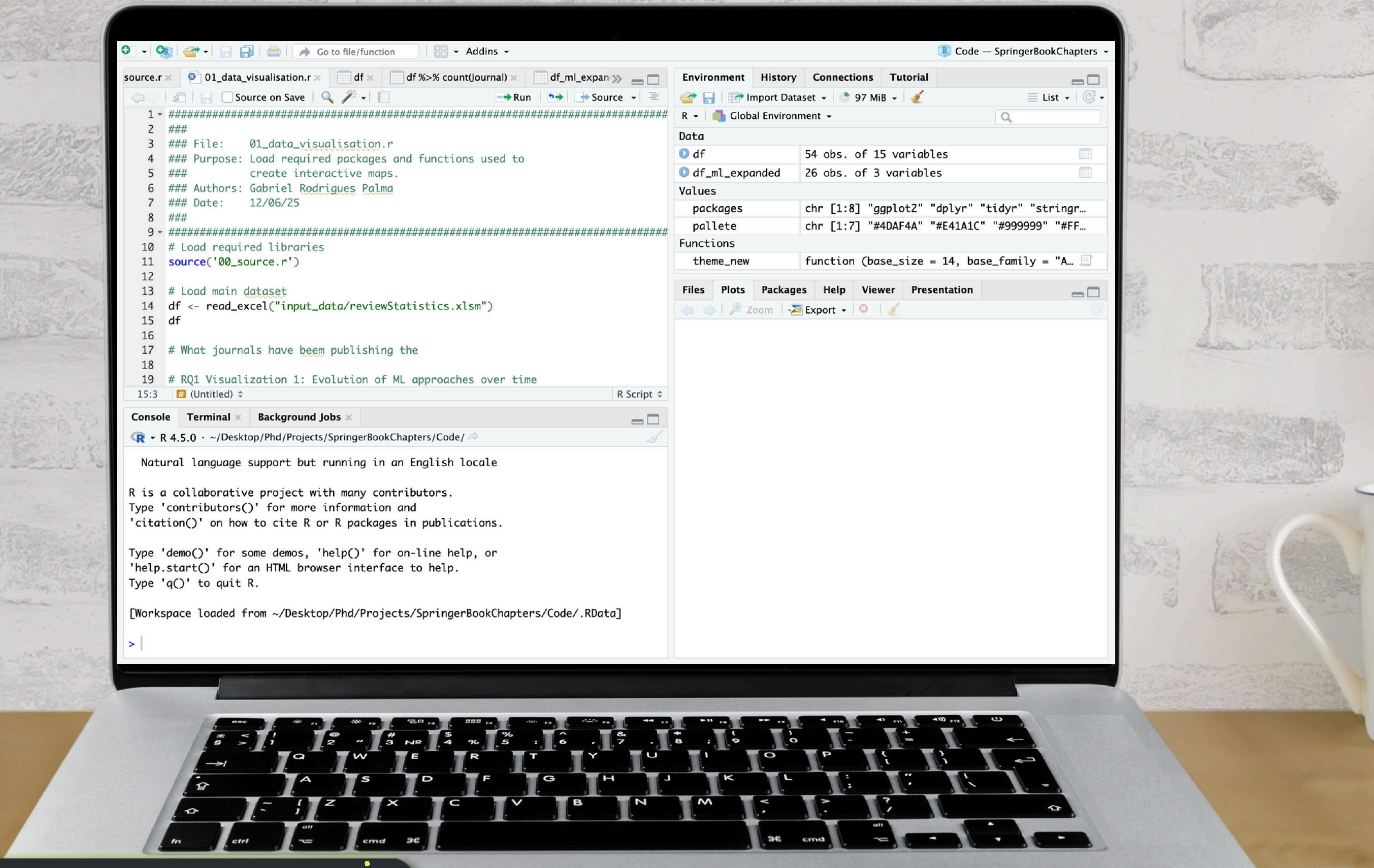
R Input

```r
# Grouped bar chart
biodiversity <- data.frame(
  habitat = c('Forest', 'Grassland'),
  native = c(45, 32),
  invasive = c(8, 15)
)
tidyr::pivot_longer(biodiversity, -habitat) %>%
  ggplot(aes(x = habitat, y = value, fill = name)) +
  geom_col(position = 'dodge')
```

→



Basic Plots



Tidyverse for Ecologists
(TIDY01)

stats

Day 2 (9:30 - 17:30)

- Data Cleaning (9:35 - 11:05)
- Data Filtering and Selecting (11:05 - 12:30)
- Grammar of Graphics (13:30 - 15:00)
- Basic Plots (15:00 - 16:30)
- Layers, Scales, and Faceting (16:30 - 17:30)

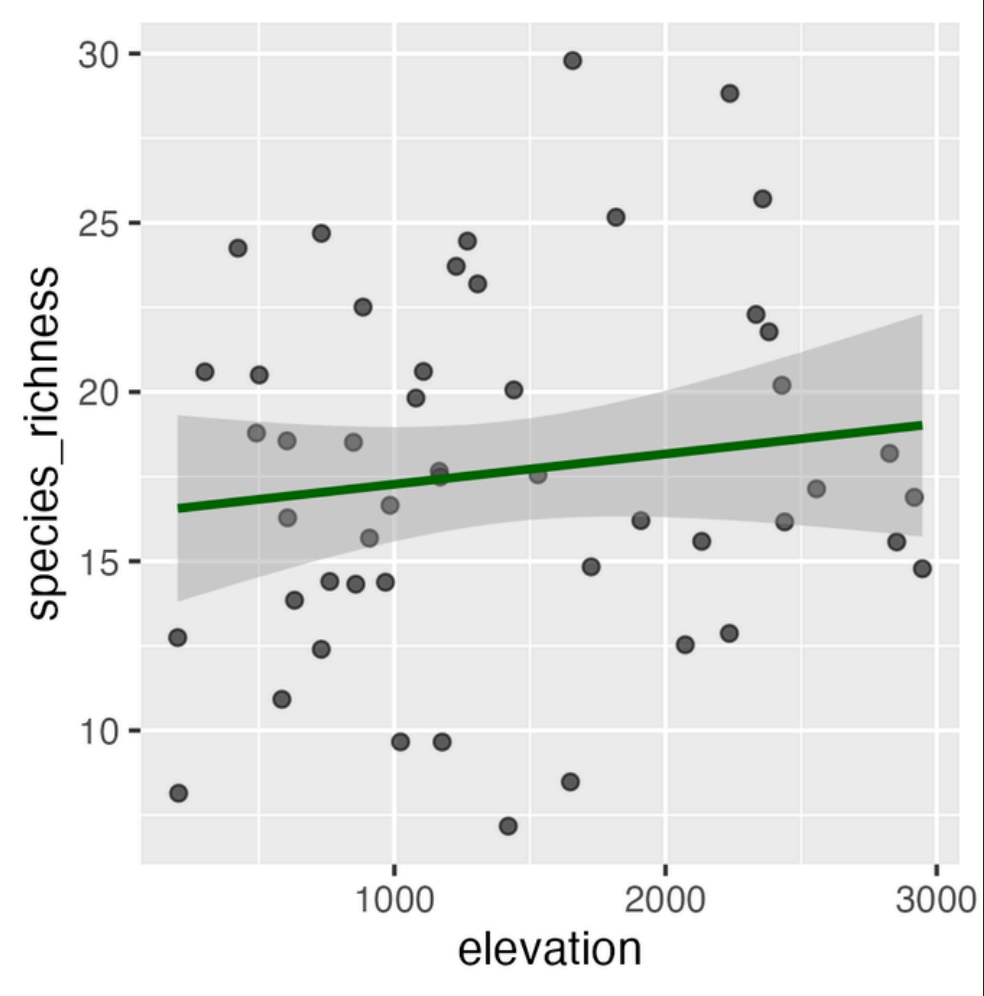
Layers, Scales, and Faceting

Layers, Scales, and Faceting

R Input

```r
# Statistical layers for biodiversity patterns
biodiversity_data <- data.frame(
  elevation = runif(50, 100, 3000),
  species_richness = 25 - 0.005 * runif(50, 100, 3000) + rnorm(50, 0, 3)
)
ggplot(biodiversity_data, aes(x = elevation, y = species_richness)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, color = "darkgreen")
```

→

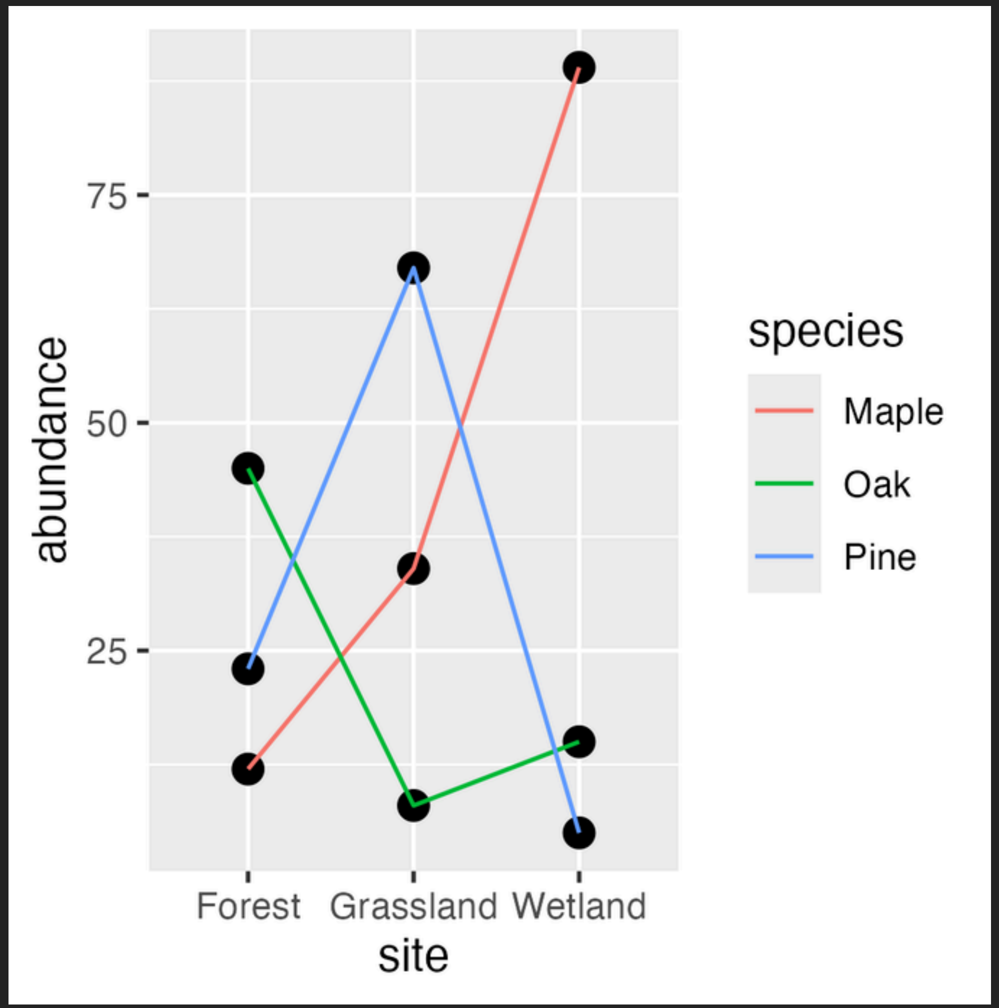


Layers, Scales, and Faceting

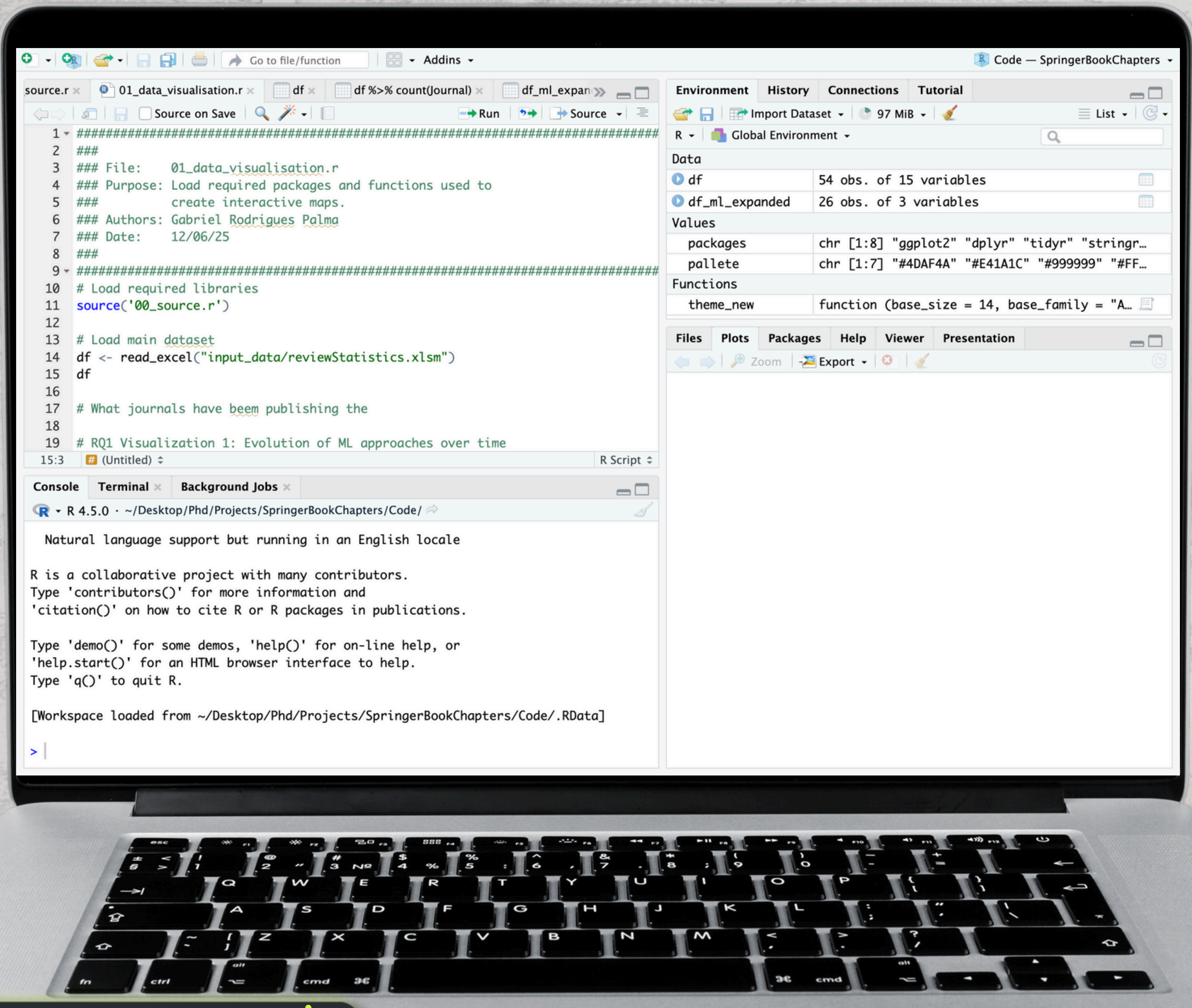
R Input

```r
# Basic layer structure with species data
library(ggplot2)
species_data <- data.frame(
  site = rep(c("Forest", "Grassland", "Wetland"), each = 3),
  species = rep(c("Oak", "Pine", "Maple"), 3),
  abundance = c(45, 23, 12, 8, 67, 34, 15, 5, 89)
)
ggplot(species_data, aes(x = site, y = abundance)) +
  geom_point(size = 3) +
  geom_line(aes(group = species, color = species))
```

→



Layers, Scales, and Faceting



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